

## Multiple Choice (10 marks)

1. If  $\mathbf{r}(t) = (6t+2)\mathbf{i} + 5t^2\mathbf{j} - 8t\mathbf{k}$ , find the Binormal vector as  $[\mathbf{x}, \mathbf{y}, \mathbf{z}]$ . What are  $x$ ,  $y$ ,  $z$ ? Return them as a list.

0.7, 0.0, 0.7

0.6, 0.0, 0.8

0.0, 0.8, 0.6

0.0, 0.7, 0.7

0.8, 0.6, 0.0

2. Use the Trapezoidal Rule with to approximate  $\int_0^\pi \sin^2(x)dx$ . Return the approximated demical value.

(a) 1.414214

(b) 1.000000

(c) 0.785398

(d) 1.570796

(e) 1.234567

3. What is the smallest whole number that has a remainder of 1 when divided by 4, a remainder of 1 when divided by 3, and a remainder of 2 when divided by 5?

(a) 72

(b) 76

(c) 67

(d) 64

(e) 57

4. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field.  $x^5 + 3x^3 + x^2 + 2x$  in  $\mathbb{Z}_5$

(a) 1,2

(b) 0,1,2,3

(c) 1

(d) 3,4

(e) 0,1,2

5. Each story of a building is 14.3 feet tall. If the height of the building is 185.9 feet, how many floors does the building have?

- (a) 10
- (b) 15
- (c) 17
- (d) 9
- (e) 11

## True / False (10 marks)

*Answer True or False*

6. True or False: The bakers can make a total of 37 apple pies using 270 apples from 15 boxes, with 7 apples required per pie.
7. True or False: A field cannot have exactly 60 elements, as the number of elements in a finite field must be a power of a prime number.
8. True or False: The volume of the solid bounded by the elliptical paraboloid and the plane is 1.5 cubic units.
9. True or False: When transforming  $\sin x$  to  $3 \sin 2x$ , the domain is halved and the range is also tripled.
10. True or False: Every field is indeed a ring, as it satisfies all ring properties including addition and multiplication.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Given that a patient takes 150 mg of a drug daily and retains 5% of the drug in their body just before the next dose, calculate the quantity of the drug present in the body after the third dose. Discuss your methodology and reasoning.
12. Evaluate the surface integral  $\iint_S \vec{F} \cdot d\vec{S}$  for the vector field  $\vec{F} = \sin(\pi x)\vec{i} + (zy^3)\vec{j} + (z^2 + 4x)\vec{k}$  over the box defined by  $-1 \leq x \leq 2$ ,  $0 \leq y \leq 1$ , and  $1 \leq z \leq 4$ . Utilize the divergence theorem, providing a detailed explanation of your steps and reasoning.

*End of Paper*